

Reports, records and a proven restore

Day 6 - Afternoon

Running scenario: Thara Hospitals patient portal pilot

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CONCEPT 1



Timed segments again

15 min

Two things govern the afternoon

- RDS takes several minutes to create and to restore. You do not watch it
- The bucket work fills the minutes while it provisions
- The database is **deleted twice** today. The snapshot is what persists

The clock starts at the failure, and stops when the rows print.

COMMON MISTAKE

Deleting an RDS instance deletes its manual snapshots.

It does not. The automated backups go. The snapshot stays.

Three things to know before you start

- A restore makes a **new** database, with a new endpoint
- The restore page offers the default network. Choose yours again
- `thara-web-1` reaches S3 and the VPC, and nothing else
- Backups go to the vault that AWS Backup offers: Default

COMMON MISTAKE

A security group on the database is the only thing keeping it private.

Placement in a private subnet does the most.

Reports, records and a proven restore

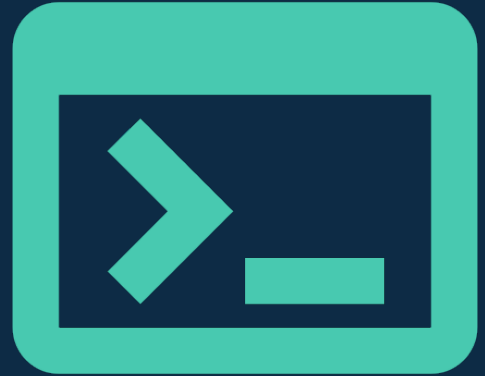
You build:

3

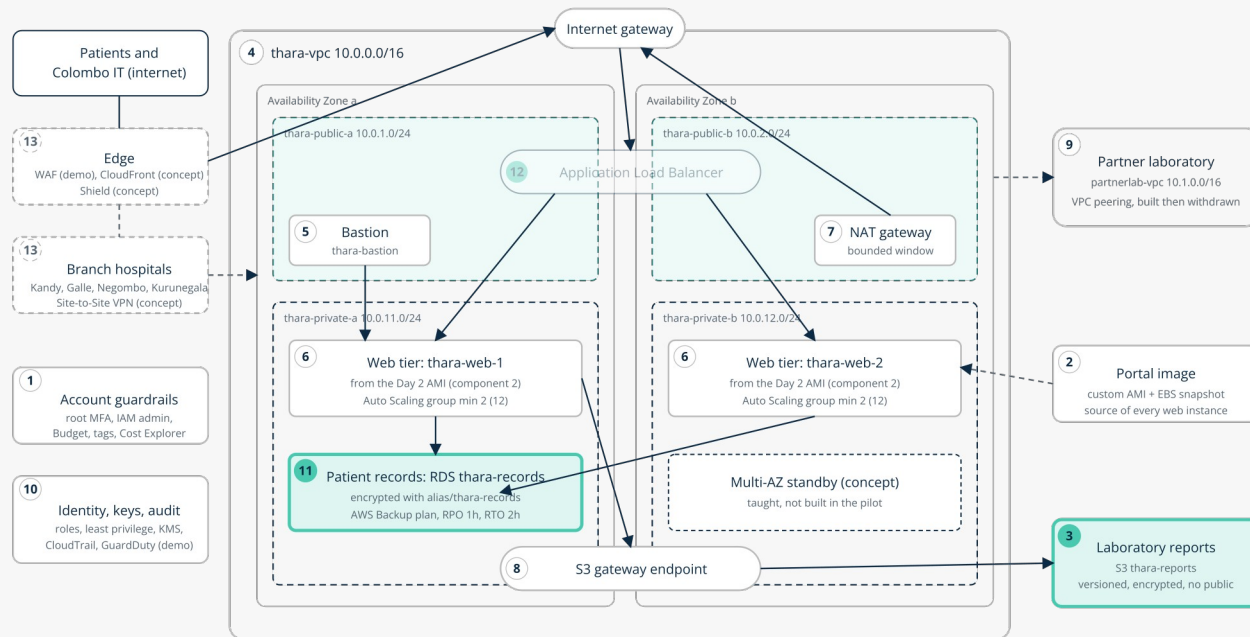
11

Estimated credit use: about USD 1.20

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The Thara architecture



TODAY

3 Laboratory reports bucket

11 Patient records database and Backup plan

KEY

Built today

Built earlier

Concept only

Not yet built

SEGMENT 1 - 35 MIN



Start the RDS create, then bucket versioning and lifecycle

Cost: db.t3.micro about USD 0.02 per hour

Start, reopen the door, place the database

1

Confirm the region is `ap-south-1` and that you are the admin user, not root.

Expected: Mumbai, and `admin-<student id>`, in the top bar.

2

Start `thara-bastion` and `thara-web-1`. Set the SSH source of `thara-bastion-sg` to My IP again. Connect, and hop to `thara-web-1`.

Expected: A prompt of the form `ec2-user@ip-10-0-11-...`

3

RDS, Subnet groups: `thara-db-subnets`, in `thara-vpc`, with the two private subnets.

Expected: Two subnets in two zones.

4

EC2, Security groups: `thara-db-sg`. One inbound rule, TCP 3306 from `thara-web-sg`.

Expected: The source is a group, not an address.

Start the database, and write the time

5 RDS, Create database: Standard create, MySQL, Free tier, `thara-records`, db.t3.micro, 20 GiB, a password of your own.

Expected: Creating. About five to ten minutes.

5 Public access No. `thara-db-subnets` and `thara-db-sg`. Encrypted with the key `thara-records`.

Expected: Go straight to step 6. Do not watch it.

Clear storage autoscaling, Extended Support and deletion protection.

While it provisions: versions and a rule

6

S3, `thara-reports-<student id>`: enable versioning. Upload a changed `reports-index.txt` from your computer. Show versions.

Expected: Two versions. The Day 2 one has the ID null.

7

Download the earlier version and upload it again. Read the file from the web tier.

Expected: Three versions. The Day 2 line prints.

8

Management, Create lifecycle rule `thara-reports-lifecycle`: Standard-IA after 90 days; noncurrent versions deleted after 180.

Expected: Enabled. Nothing moves today.

SEGMENT 2 - 20 MIN



Bucket policy scoped to the role; confirm Block Public Access

Cost: nil

The bucket says who may read it



9

Permissions, Bucket policy, Edit, Add new statement. Keep the Version line. Two statements that name `thara-web-role`. List from `thara-web-1`.

Expected: Saved. The listing still works.

10

Stay on Permissions. Read the two lines about public access.

Expected: Block all public access: On. Not public.

The policy: one principal, two actions



```
{  "Effect": "Allow",
  "Principal": {  "AWS":
    "arn:aws:iam::<account id>:role/thara-web-role"  },
  "Action": "s3:ListBucket",
  "Resource": "arn:aws:s3:::thara-reports-<student id>" }
```

The second statement allows s3:GetObject on the objects: the same resource with /*.

SEGMENT 3 - 35 MIN



Connect to RDS, create a table, snapshot, delete

Cost: nil

Connect, and put three rows in it

11 When `thara-records` is Available, write the time and copy its endpoint. On `thara-web-1`, install the client and connect.

Expected: Complete! Then a prompt that ends >

12 Create the database `thara` and the table `patients`, insert three test rows, and select them.

Expected: Three rows. Test names only.

```
$ sudo dnf install -y mariadb105
$ mysql -h <endpoint> -u admin -p --ssl
```

Snapshot, then delete on purpose



13

Actions, Take snapshot: `thara-records-snap1`. Wait for Available on the Manual tab.

Expected: Encrypted with the key. The System tab shows RDS's own backup.

14

Delete `thara-records`. Clear the final snapshot and Retain automated backups. Type delete me.

Expected: The System tab empties. `thara-records-snap1` remains.

Write the time of the deletion. The clock for your RTO starts there.

SEGMENT 4 - 35 MIN



Restore from snapshot; measure RTO; delete again

Cost: nil

Restore into the right place

15 Snapshots, Manual, Restore snapshot: `thara-records-r1`, `db.t3.micro`, `thara-db-subnets`, Public access No, `thara-db-sg`. Write the time.

Expected: Creating. The page offered the default group.

16 When it is Available, write the time. Copy the new endpoint and select the rows. Write the time again.

Expected: The same three rows. Deletion to rows is your measured RTO.

```
$ mysql -h <new endpoint> -u admin -p --ssl \  
$ -e "SELECT * FROM thara.patients;"
```

Break it now, then delete the copy



17

The break-it needs a database. Do it now, then come back.

Expected: Your laptop cannot connect, with the group wide open.

18

Delete `thara-records-r1` in the same way. Keep `thara-records-snap1`.

Expected: No database exists. The snapshot is listed.

Nothing bills by the hour once step 18 is done. Do not delete the snapshot.

SEGMENT 5 - 25 MIN

AWS Backup plan on an EBS volume; restore; measure

Cost: negligible

A plan, and what it covers

19 EC2, Snapshots, the Day 2 data volume: Create volume from snapshot, gp3, 8 GiB, zone a, with Day=6.

Expected: Available. Copy its id. Do not attach it.

20 AWS Backup, Build a new plan: `thara-daily`, rule `thara-daily`, vault Default, Daily, 7 days. Assign EBS by the tags Module=CC and Day=6.

Expected: One rule and one resource assignment.

Two tags mean both. Module=CC alone would match everything you have tagged.

Back up now, restore, read

21

Protected resources, Create on-demand backup: EBS, the volume, now, 7 days, Default.

Expected: The backup job shows Completed.

22

Choose the recovery point, Restore: EBS volume, gp3, 8 GiB, zone a. Attach the new volume to `thara-web-1` as `/dev/sdf`. Mount and read.

Expected: The restore job shows Completed. The Day 2 line prints.

```
$ sudo mkdir -p /data
$ sudo mount /dev/nvme1n1 /data
$ cat /data/reports-index.txt
```

SEGMENT 6 - 15 MIN



Ledger, teardown, evidence

Cost: nil

Prove it, then break it



PROVE IT

- Two versions of reports-index.txt, and the earlier one current again
- The three patient rows, after the restore from `thara-records-snap1`
- The Backup restore job Completed, and the file read on the restored volume

BREAK IT

- Open `thara-db-sg` to 0.0.0.0/0 on port 3306
- From your laptop, test TCP 3306 to the endpoint. It still fails
- nslookup the endpoint: a private address. Explain, then revert

Tear down and record



TEARDOWN

- No RDS instance exists. `thara-records-snap1` remains
- Delete the restored volume and the temporary volume. Keep the plan `thara-daily`
- Stop `thara-bastion` and `thara-web-1`

COST

Estimated credit use: about USD 1.20

FOR YOUR EVIDENCE PACK

- Version list of `reports-index.txt` showing the restore
- Lifecycle rule and bucket policy screenshots; Block Public Access shown on
- Ledger rows with the RDS create time, delete time, restore time and the measured RTO; Backup restore time
- Screenshot of the three rows after restore

Common mistakes



COMMON MISTAKE

Deleting an RDS instance deletes its manual snapshots.

It does not. A manual snapshot stays until you delete it.

COMMON MISTAKE

A security group on the database is the only thing keeping it private.

Placement in a private subnet does the most.

Before the next session



READ

Well-Architected Framework

The summary of the Reliability pillar. Also Wittig and Wittig, the chapters on Auto Scaling and load balancing.

SELF-CHECK

CLF-C02 bank

Domain 1 set B, 10 items.

PQ6 is due.

TASK

Your group

Complete the capstone data tier in the host account.

Questions & Discussion



Next: high availability, delivered and monitored.



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